PART A: RISK MANAGEMENT

🔹 Definition (2 marks)

Risk management = systematic process to identify, analyse, and mitigate risks in IT systems □

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🔹 CIA TRIAD (VERY IMPORTANT – 4 marks)

Confidentiality → Only authorised access

Integrity → Data accuracy & completeness

Availability → Access when needed

👉 Core of cybersecurity □

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🔹 KEY TERMS (4–6 marks)

Asset → Anything valuable (data, systems, hardware)

Threat → Potential harmful event (hackers, disasters)

Vulnerability → Weakness exploited by threat

Risk = Threat × Vulnerability × Impact

🔹 TYPES OF ASSETS (4 marks)

Physical → servers, devices

Software → apps, OS

Information → customer data, records

👉 Includes financial + reputational value □

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🔹 TYPES OF THREATS (4–6 marks)

Physical → theft, damage

Technical → malware, hacking

Human → errors, insider threats

Environmental → fire, power failure

🔹 RISK MANAGEMENT PROCESS (6–10 marks ⭐)

Identify risks

Analyse threats & vulnerabilities

Assess likelihood & impact

Prioritise risks

Implement mitigation strategies

Monitor continuously

👉 Ensures security + business continuity □

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🔹 RISK MITIGATION (EXTRA POINT ⭐)

Avoid risk

Reduce risk

Transfer risk (insurance)

Accept risk

🪚 PART B: INDUSTRY STANDARDS

🔹 Definition (2 marks)

Industry standards = agreed norms/specifications guiding technology and practices □

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🔹 WHY IMPORTANT (6–10 marks ⭐)

Quality assurance → reliable systems

Interoperability → systems work together

Safety & security → reduce risks

Global trade → compatibility across countries

Compliance → legal & audit requirements

👉 Key exam explanation point □

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🔹 WITHOUT STANDARDS (4–6 marks)

Incompatible systems

Safety risks

Higher costs

Poor quality

Limited global trade

More e-waste

🔹 REAL CASE STUDY (VERY IMPORTANT ⭐)

🚀 Mars Climate Orbiter (10 marks example)

Failure due to metric vs imperial unit mismatch

Result: spacecraft destroyed

Lesson: need for standardisation across teams □

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🔹 ORGANISATIONS (2–4 marks)

IEEE → engineering standards

ISO → international standards

ACM → ethical guidelines